

ω Cen: orbital histories of the progenitor dwarf

Three classes of initial conditions for the host-disruption simulations

oCen_dm project notes

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Abstract

The mass-modelling track infers ω Cen’s dark-matter content from its present-day kinematics. This note prepares the complementary track: simulate the tidal disruption of the host dwarf and ask how much dark matter the surviving nucleus is left with. A disruption simulation needs an orbital history, and three classes were defined: (1) the present orbit in a static axisymmetric potential; (2) the present orbit integrated backwards with dynamical friction, so the dwarf arrives from the outer halo and sinks; (3) deposition by the Gaia–Sausage–Enceladus (GSE) merger followed by inward migration driven by the decelerating Galactic bar (Dillamore et al., 2026). For each class we state the model, examine the assumptions that matter, and pick three representative orbits; the nine orbits are then compared through per-Gyr pericentres, apocentres, tidal-field strength and Jacobi radii. The main findings: class 2 is controlled by the satellite’s mass-loss history rather than its mass, and the selected histories converge to today’s orbit over the last 3–5 Gyr; class 3 reproduces the published result that migration from GSE debris requires a present-day pattern speed $\Omega_b \lesssim 26 \text{ km s}^{-1} \text{ kpc}^{-1}$, places the nucleus on a most-bound GSE-debris orbit (apocentre 11–12 kpc, pericentre 0.4 kpc) until ~ 2.5 Gyr ago, and shows that a slow bar also alters “today’s orbit” (pericentre 0.8 rather than 1.6 kpc). Constant-mass friction experiments show a substantial change between 10^7 and $10^8 M_\odot$ and no GSE overlap for the tested $10^9 M_\odot$ case. This motivates early envelope loss but does not establish a universal mass bound. The classes differ most in the pericentric tides 5–10 Gyr ago, by more than an order of magnitude.

1 Purpose and scope

The question of the project is whether ω Cen is embedded in dark matter. The kinematic route constrains the mass profile today. The dynamical route asks what a disrupting dwarf leaves behind, which depends on how the dwarf was stripped, hence on its orbit through the Milky Way. Before running any N -body model we fix a small set of orbital histories that span the plausible range. This note is about those histories only; no disruption has been simulated.

All code is in `src/ocen_dm/tails/`: `progenitor_orbits.py` (classes 1 and 2, the orbit set), `bar_migration.py` (class 3), `products.py` and `writeup_figures.py` (tables and figures). Tests: `tests/test_progenitor_orbits.py`, `tests/test_bar_migration.py` (the original validation was recorded in the journal). Numerical products are in `results/tails/`.

2 Present-day phase-space point

The adopted centre from `cluster.py` is $\alpha = 201.69683^\circ$, $\delta = -47.47957^\circ$. All classes use distance $D = 5.43 \pm 0.05 \text{ kpc}$, systemic proper motion $(\mu_{\alpha*}, \mu_\delta) = (-3.257, -6.730) \pm 0.025 \text{ mas yr}^{-1}$ (Vasiliev & Baumgardt, 2021), line-of-sight velocity $232.7 \pm 0.21 \text{ km s}^{-1}$ (Baumgardt & Vasiliev, 2021). Classes 1 and 2 use the solar parameters of the McMillan (2017) potential, $R_0 = 8.21 \text{ kpc}$, $v_0 = 233.1 \text{ km s}^{-1}$, $z_\odot = 20.8 \text{ pc}$, $(U, V, W)_\odot = (11.1, 12.24, 7.25) \text{ km s}^{-1}$ (Schönrich et al., 2010); class 3 follows Dillamore et al. (2026) and uses the `astropy` default frame ($R_0 = 8.122 \text{ kpc}$, $\mathbf{v}_\odot = (12.9, 245.6, 7.78) \text{ km s}^{-1}$).

Table 1: Present-day galactocentric state (McMillan 2017 frame). L_z is prograde-positive throughout this [note](#).

(x, y, z)	$(-4.90, -4.07, 1.41)$ kpc, $r = 6.52$ kpc
(v_x, v_y, v_z)	$(95.2, -28.9, -87.1)$ km s $^{-1}$
L_z	-529 kpc km s $^{-1}$ (retrograde)
E (McMillan 2017)	-1.85×10^5 km 2 s $^{-2}$
E (Hunter et al. 2024, axisymmetrised)	-1.46×10^5 km 2 s $^{-2}$

A frame error was caught and is now guarded by a test: `galpy`’s `solarmotion` is the peculiar solar motion alone, and adding v_0 to it produced a prograde orbit with apocentre 10.6 kpc. The corrected conversion agrees with an independent `astropy` transform to 0.01 kpc and 0.1 km s $^{-1}$.

3 Class 1: today’s orbit in a static axisymmetric potential

3.1 Construction

The present-day state is integrated backwards for 10 Gyr with `galpy` (`dop853_c`, [Bovy, 2015](#)) in three published potentials: McMillan (2017), the `galpy` `MWPotential2014`, and model I of [Irrgang et al. \(2013\)](#). No friction, no mass loss, no time dependence: the dwarf is assumed to have been on this orbit throughout.

3.2 Assumptions

The only assumption is the potential. The three chosen bracket the range of circular-speed curves and bulge/disc masses in common use; the potentials’ own solar parameters are used (Irrgang: $R_0 = 8.4$ kpc, $v_0 = 242$ km s $^{-1}$).

3.3 Result

Table 2: Class-1 orbits (kpc). McMillan 2017 reproduces the AGAMA orbit computed on 2026-09-19 for the Jacobi radius (1.59/6.99); the two implementations differ by 4% in pericentre.

potential	r_{peri}	r_{apo}	e	z_{max}
McMillan 2017	1.57	7.04	0.63	2.88
MWPotential2014	1.97	6.76	0.55	2.72
Irrgang 2013 model I	1.25	7.20	0.70	3.06

The pericentre varies by a factor 1.6 between potentials (Fig. 1); the apocentre by 6%. All three are kept: the class-1 spread is the potential spread.

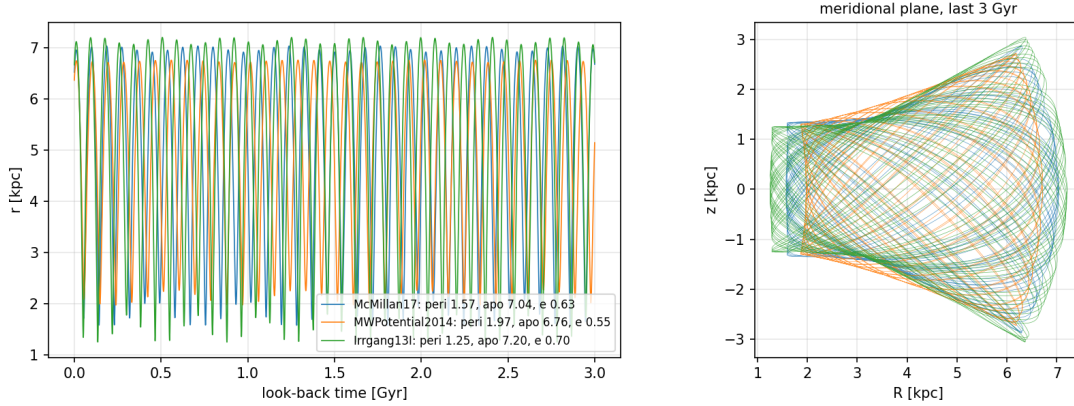


Figure 1: Class 1. Left: galactocentric radius over the last 3 Gyr in the three potentials. Right: the same orbits in the meridional plane. The orbit is a thick, eccentric, retrograde loop confined to $z < 3$ kpc.

4 Class 2: today’s orbit with dynamical friction integrated backwards

4.1 Construction

The equation of motion is integrated backwards in time with the Chandrasekhar deceleration (Chandrasekhar, 1943) added,

$$\mathbf{a}_{\text{df}} = -4\pi G^2 M(t) \rho(r) \ln \Lambda \left[\text{erf}(X) - \frac{2X}{\sqrt{\pi}} e^{-X^2} \right] \frac{\mathbf{v}}{v^3}, \quad (1)$$

$$X = \frac{v}{\sqrt{2}\sigma}, \quad \sigma = \frac{v_c(r)}{\sqrt{2}}, \quad \ln \Lambda = \ln \frac{r}{\max(r_h, GM/v^2)}. \quad (2)$$

which is `galpy`’s prescription with its default Coulomb logarithm. Reversing time reverses the sign of the friction term, so the orbit is pumped outwards: the integration recovers where a satellite of bound mass $M(t)$ must have been to sink to the present orbit. The host is McMillan (2017) evaluated with AGAMA (Vasiliev, 2019); the integrator is a leapfrog with 0.25 Myr steps (about 1 s per 10 Gyr orbit). The frictionless run reproduces class 1 to four figures (1.573/7.037 versus `agama.orbit` 1.572/7.037 kpc in the same potential), and a constant-mass case checked against `galpy`’s `ChandrasekharDynamicalFrictionForce` agrees to 0.1% in pericentre and 4% in apocentre (1.786/44.3 versus 1.786/46.0 kpc; 1 s versus 482 s). (A first version of the integrator converted accelerations with the wrong power of the time unit, making kicks 4.4% too weak; Codex caught it on 2026-09-20 and every leapfrog result in this note was regenerated.)

4.2 Assumptions, and which one matters

Friction scales with the satellite’s bound mass, so the backward orbit depends on the *function* $M(t)$. Integrating with a constant mass is not a conservative simplification: it is wrong in the direction that matters. A constant $10^{10} M_\odot$ satellite is pumped to 40–140 kpc within 10 Gyr, i.e. it “arrives” from outside the virial radius although the real object had already been stripped to a small fraction of that mass for most of its orbit (Fig. 2, right).

We therefore parametrise the history and let a physical requirement select it:

1. **Infall mass** M_{inf} . The nuclear-cluster–host relation gives $M_\star \approx 10^9 M_\odot$ for the host (Pfeffer et al., 2021); oMEGACat X quotes a “GE-like” dwarf mass $4.5^{+5.7}_{-2.4} \times 10^9 M_\odot$ and favours

Sequoia/Thamnos over GSE as the debris (Souza et al., 2026). Halo masses for $M_\star = 10^8$ – $10^9 M_\odot$ at $z \sim 2$ are 10^{10} – $10^{11} M_\odot$; GSE itself is $2 \times 10^{11} M_\odot$ (Naidu et al., 2021). Scanned: 10^{10} , 3×10^{10} , 10^{11} , $2 \times 10^{11} M_\odot$.

2. **Infall time** $t_{\text{inf}} = 10$ Gyr ago ($z \approx 2$; the GE merger is dated 8–11 Gyr ago). Also scanned 8 Gyr; same conclusions.
3. **Stripping law** $M(t) = M_{\text{inf}} \exp[-(t_{\text{inf}} - t_{\text{lb}})/\tau]$, floored at the nucleus mass $3.55 \times 10^6 M_\odot$ (Baumgardt & Hilker, 2018); τ scanned 0.5–3 Gyr. Half-mass radius $r_h = 1 \text{ kpc} (M_{\text{inf}}/10^{10} M_\odot)^{1/3}$, fixed.
4. **Selection rule.** The orbit must be at the virial radius of the young Milky Way, 50–150 kpc, at t_{inf} . Histories that do not reach 50 kpc could not have sunk to today’s orbit by friction alone; histories beyond 150 kpc would need more friction than the mass allows.

4.3 Scan and picks

Figure 3 shows the apocentre in the last Gyr before infall over the (M_{inf}, τ) grid. The table is noisy at the 20% level because the phase of the last apocentre matters, but the trends are clear: a Sequoia-scale $10^{10} M_\odot$ halo reaches the virial radius only with slow stripping ($\tau \gtrsim 1.5$ Gyr); a GSE-scale halo must be stripped fast ($\tau \lesssim 1$ Gyr) or it arrives from far outside the virial radius. Three histories spanning the allowed band are kept (Table 3). In every viable history the bound mass is below $10^9 M_\odot$ by 5 Gyr ago and the orbit has today’s shape for the last 3–5 Gyr (Fig. 2, middle): *the present orbit is a good proxy for the second half of the history in all cases*, and the classes differ in the first half.

Table 3: Class-2 picks (kpc). Columns 5–6: pericentre–apocentre range in the stated look-back window.

label	$M_{\text{inf}} [M_\odot]$	τ [Gyr]	r at infall	5–6 Gyr ago	2–3 Gyr ago
Sequoia-like	10^{10}	2	65	3.0–18.1	1.9–9.2
intermediate	3×10^{10}	1.2	86	2.2–11.7	1.6–7.3
GSE-like	10^{11}	0.75	77	1.7–7.4	1.6–7.0

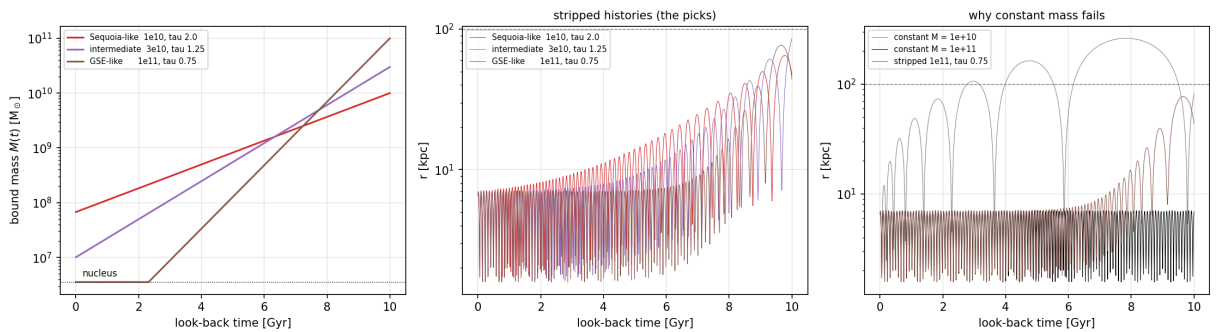


Figure 2: Class 2. Left: the three adopted bound-mass histories. Middle: the corresponding backward orbits; the dashed line marks 100 kpc. Right: why a constant mass fails — a constant $10^{10} M_\odot$ satellite (grey) is pumped beyond the virial radius within 10 Gyr, a constant $10^{11} M_\odot$ one (black) escapes even faster, while the stripped 10^{11} history (brown) stays on today’s orbit for 6 Gyr and arrives from 110 kpc.

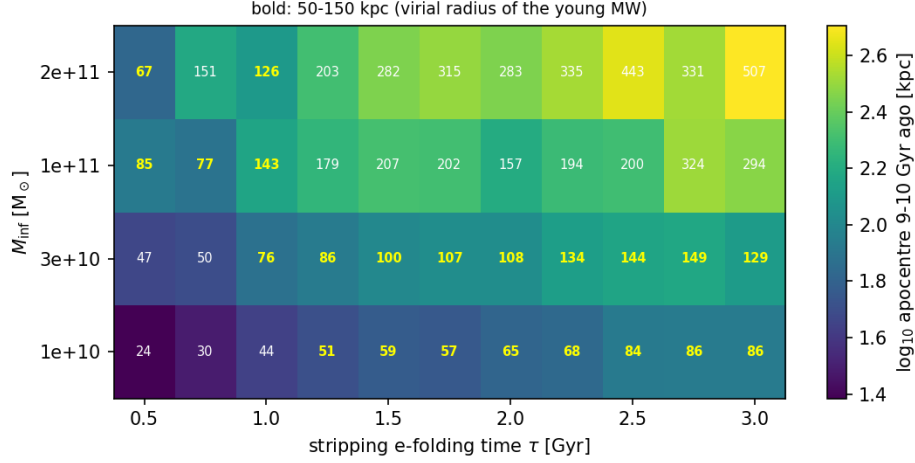


Figure 3: Class 2: apocentre 9–10 Gyr ago as a function of infall mass and stripping e-folding time ($t_{\text{inf}} = 10$ Gyr). Bold entries satisfy the 50–150 kpc selection rule. Low-mass halos need slow stripping, GSE-mass halos fast stripping.

5 Class 3: deposited by GSE, migrated inward by the bar

5.1 What the literature provides

Dillamore et al. (2026) showed that ω Cen can be traced back to the GSE debris region of (L_z, E) if the Galactic bar has decelerated and its present pattern speed is $\Omega_b \lesssim 26 \text{ km s}^{-1} \text{ kpc}^{-1}$, well below most current estimates. The mechanism is the retrograde 1:1 resonance, $\Omega_\phi - \Omega_b + \Omega_r = 0$, which scatters orbits to lower E and more retrograde L_z . Friction on the nucleus is negligible ($\dot{L}_z \approx 9.6 \text{ kpc km s}^{-1} \text{ Gyr}^{-1}$). The GSE debris itself is characterised by Belokurov et al. (2023): $|L_z| < 700 \text{ kpc km s}^{-1}$, chevrons with apocentres 11.5–25 kpc, and a set of (L_z, E) contours that we use directly.

5.2 Construction: reproducing the published set-up

The authors’ pipeline (repository `adllmr/oCen_bar`, local clone `~/Work/Code/oCen_bar`) was followed step by step; its potential files and contour file are referenced, not copied.

- **Potential.** Hunter et al. (2024) Milky Way with the Sormani et al. (2022) bar ($M_{\text{bar}} = 1.83 \times 10^{10} M_\odot$), as AGAMA files. The model is decomposed as $\Phi = \Phi_{\text{axi}} + A(t) \Phi_{\text{bar}}[S(t)] - A(t) \bar{\Phi}_{\text{bar}}[S(t)]$, i.e. the axisymmetric total plus the scaled, rotating baryonic bar part minus the same part azimuthally averaged, so the $m = 0$ mass is unchanged while the bar grows.
- **Bar history.** Amplitude A follows the Dehnen (2000) switch-on between $t = 0$ and $t_1 = 1$ Gyr. The pattern speed is Ω_1 until t_1 , starts decelerating smoothly to $t_2 = 2$ Gyr, then obeys $\eta \equiv -\dot{\Omega}_b/\Omega_b^2 = 0.003$ until $t_f = 8$ Gyr (the bar age). Ω_1 is fixed by the present-day Ω_b : $\Omega_1 = 45.1 \text{ km s}^{-1} \text{ kpc}^{-1}$ for $\Omega_b = 24$ (the paper quotes “ ≈ 45 ”, which confirms $t_1, t_2 = 1, 2$ Gyr rather than the 2, 3 defaults left in one of the repository functions), 72 for 30, 110 for 35. The bar length scales as $S(t) = \Omega_b(t_f)/\Omega_b(t)$, approximately the corotation radius, at fixed amplitude (Fig. 6, left).
- **Samples.** $N = 10^3$ draws of the present-day state (Sec. 2), bar angle 28° , integrated from t_f back to 0 with `agama.orbit`.
- **Time unit.** All class-3 times follow the paper’s convention and are in $\text{kpc}/(\text{km s}^{-1}) = 0.978 \text{ Gyr}$, so “8 Gyr” means $8 \times 0.978 = 7.8 \text{ Gyr}$ and η is per that unit; the labels in this section keep the paper’s rounding.

- **Test.** (E, L_z) at $t = 0$ in the axisymmetric potential, compared with the outermost Belokurov et al. (2023) contour after matching the energy zero point at $R = 8.2$ kpc. Our success metric is the fraction of samples inside that contour.

5.3 Results

Table 4: Fraction of the 10^3 samples inside the outer GSE contour 8 Gyr ago.

Ω_b [km s ⁻¹ kpc ⁻¹]	20	21	22	22.5	23	23.5	24	24.5	25	25.5	26	26.5	≥ 27
fraction	0.44	0.38	0.58	0.85	0.01	0.34	0.89	0.81	0.79	0.54	0.18	0.02	0.00

- The published result is reproduced: overlap with GSE only for $\Omega_b \lesssim 26$, none for the mainstream 30–40 km s⁻¹ kpc⁻¹ (Fig. 4).
- Cross-check against the authors’ stored grid (`oCen_bar/artifacts/03_orbit_grid.npz`): 0.87, 0.69, 0.08 at $\Omega_b = 24, 25, 26$ with a dip at 23 — the same pattern. The dip is stable against the random seed (0.013 versus 0.028) and against the bar timings: it is resonant-phase sensitivity, not noise. The Ω_b dependence is intrinsically jagged.
- **Migration is late.** E and L_z of the successful samples stay at GSE-like values from 8 to ~ 2.5 Gyr ago and move to today’s values only in the last ~ 2.5 Gyr, while Ω_b falls from ~ 30 to 24 and the resonance sweeps across ω Cen (Fig. 5). The pre-migration L_z is *slightly prograde to zero* (+70 to +230 kpc km s⁻¹), not retrograde.
- **The bar changes today’s orbit.** At $\Omega_b = 24$ corotation is at ≈ 9.5 kpc, so ω Cen at 6.5 kpc is inside corotation. Its present orbit in the barred potential has peri/apo = 0.81/9.3 kpc over the last Gyr, versus 1.56/7.02 in the axisymmetrised Hunter potential and 1.57/7.04 in McMillan 2017. “Today’s orbit” in class 1 is therefore conditional on the bar being fast or absent.

Table 5: Class-3 picks at $\Omega_b = 24$: samples at the 16/50/84th percentiles of $E(t = 0)$ among those ending inside GSE. Elements from a 0.25-Myr re-integration (the 20-Myr grid cadence overestimates pericentres by up to a factor 2, since a passage at 0.4 kpc lasts ~ 1 Myr).

sample	$E(0)$ [10 ⁵]	$L_z(0)$	pre-migration peri/apo/ $e/z_{\max}$ (7–8 Gyr ago)	last Gyr peri/apo
986	−1.301	+69	0.40 / 10.9 / 0.93 / 4.8	0.84 / 9.1
133	−1.265	+163	0.42 / 11.6 / 0.93 / 4.9	0.80 / 9.3
240	−1.233	+229	0.45 / 12.2 / 0.93 / 5.2	0.84 / 9.1

5.4 Does a massive remnant migrate like a test particle?

The back-integration treats ω Cen as a test particle. The bound system whose dark-matter content we want to know is nucleus + remnant halo, and the one process that distinguishes it from a particle at leading order is dynamical friction, which scales with its mass. We repeated the class-3 integration with Chandrasekhar friction for a constant bound mass M (a controlled sensitivity experiment, not a bound on an unspecified mass-loss history), evaluated with the time-dependent, bar-inclusive density, $\sigma = v_c/\sqrt{2}$ of the axisymmetric potential and $r_h = 0.1$ kpc $(M/10^8 M_\odot)^{1/3}$; 100 samples, vectorised leapfrog with 0.25 Myr steps in the AGAMA potential (`tails/friction_test.py`). At the nucleus mass the implementation gives $\dot{L}_z = 11$ kpc km s⁻¹ Gyr⁻¹ and $\dot{E} = 7 \times 10^2$ km² s⁻² Gyr⁻¹, matching

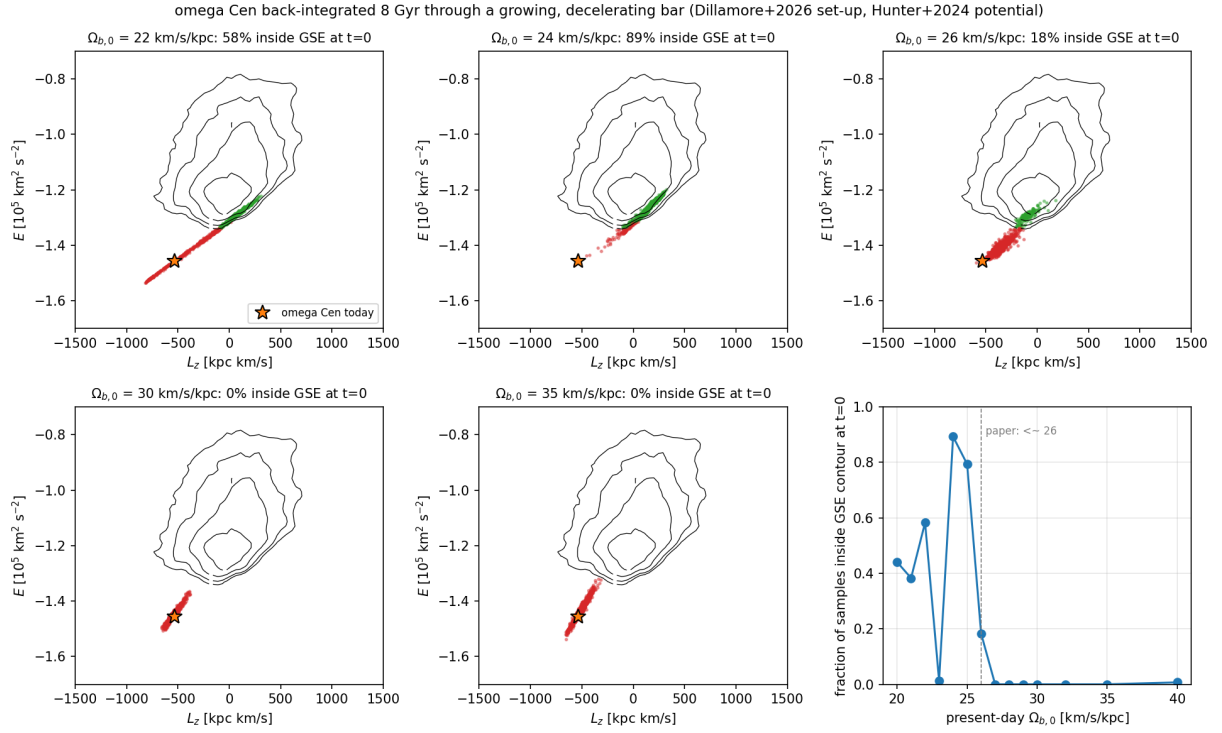


Figure 4: Class 3, the (L_z, E) plane in the axisymmetrised Hunter et al. (2024) potential. Black: GSE contours of Belokurov et al. (2023). Star: ω Cen today. Points: the 10^3 samples 8 Gyr ago after back-integration through the decelerating bar, green inside / red outside the outer contour. Bottom right: fraction inside versus present-day pattern speed; the dashed line is the paper’s $\Omega_b \lesssim 26$.

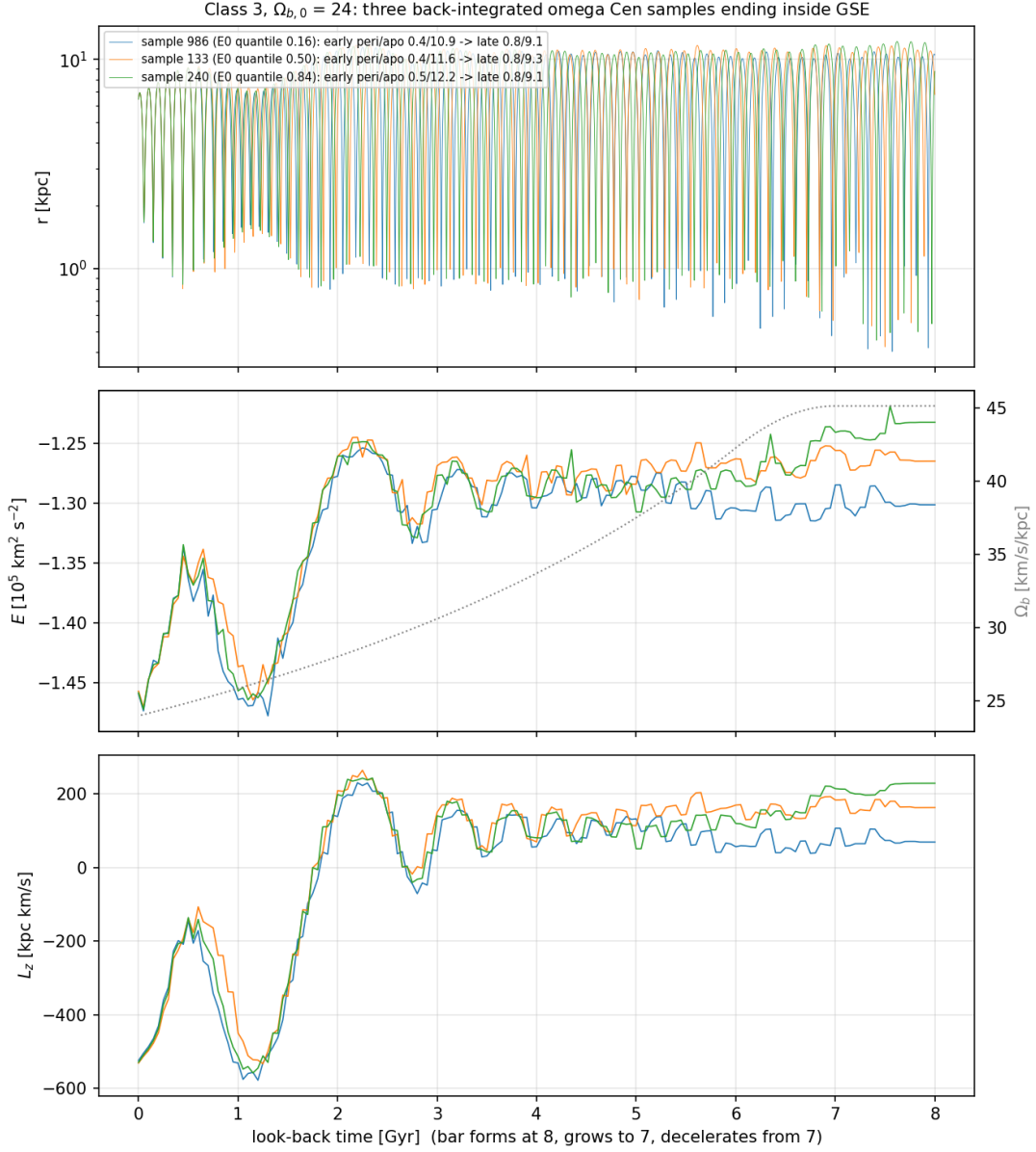


Figure 5: Class 3, the three picked samples at $\Omega_b = 24$: radius, energy (dotted grey: $\Omega_b(t)$) and L_z against look-back time. The bar forms 8 Gyr ago and grows until 7; the nucleus sits on a GSE-debris orbit until ~ 2.5 Gyr ago and is then carried to today's (E, L_z) by the retrograde 1:1 resonance. Look-back time here is $t_f - t$.

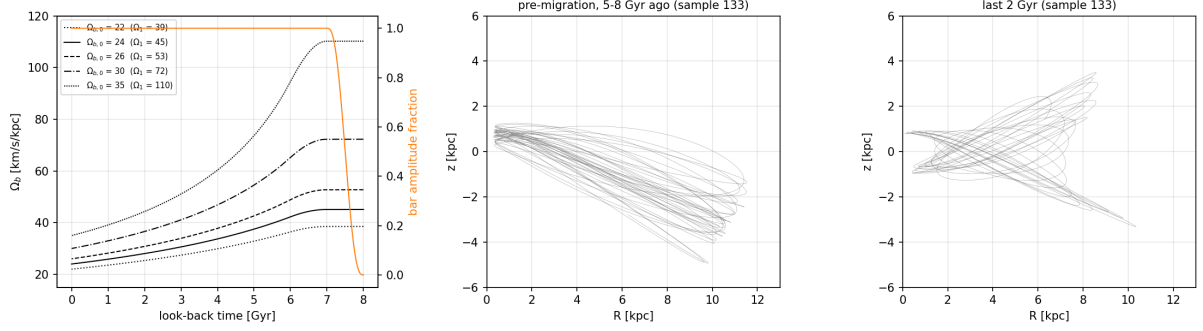


Figure 6: Class 3. Left: pattern-speed histories for several present-day values (black) and the bar amplitude switch-on (orange, right axis); with fixed $\eta = 0.003$ a present-day $\Omega_b = 35$ demands an initial $\Omega_1 = 110 \text{ km s}^{-1} \text{ kpc}^{-1}$. Middle and right: the median pick (sample 133) in the meridional plane before migration and in the last 2 Gyr — a plunging, highly inclined debris orbit turning into a rounder, lower orbit.

Table 6: Fraction of samples inside the outer GSE contour 8 Gyr ago with friction for a constant bound mass. Test-particle values from Table 4 for comparison.

Ω_b [km s $^{-1}$ kpc $^{-1}$]	20	21	22	22.5	23	24	25	26	27
test particle	0.44	0.38	0.58	0.85	0.01	0.89	0.79	0.18	0.00
$M = 10^7 M_\odot$	0.81	0.55	0.77	0.02	0.89	0.93	0.98	0.64	0.03
$M = 10^8 M_\odot$	0.00	0.07	0.47	0.15	0.44	0.34	0.00	0.00	0.00
$M = 10^9 M_\odot$	0.00 at $\Omega_b = 24$; orbit pumped to a 79 kpc apocentre								

the paper’s 9.6 and 6.3×10^2 ; the frictionless leapfrog reproduces `agama.orbit` (0.87 versus 0.89 inside GSE).

The answer is mass-dependent, with the transition between 10^7 and $10^8 M_\odot$:

- $M = 10^7 M_\odot$: high GSE-overlap fractions at several pattern speeds, but a strong dip at $\Omega_b = 22.5$. Friction changes the resonance-sensitive window rather than preserving the test-particle grid point by point.
- $M = 10^8 M_\odot$: marginal. At $\Omega_b = 22$ –24 a third to a half of the samples still end inside GSE, none for $\Omega_b \geq 25$; the resonant and friction torques are comparable and the endpoint is pushed to $E \approx -0.8 \times 10^5$, the upper edge of the debris.
- $M \geq 10^9 M_\odot$: friction dominates; the system would have come from the outer halo (class 2 territory), not from GSE debris.

These constant-mass experiments place the transition in sensitivity between 10^7 and $10^8 M_\odot$ for the chosen bar histories; they suggest substantial envelope loss before migration. They do not define a universal $10^8 M_\odot$ upper limit. A real mass history can exert more or less accumulated drag than a fixed-mass trial, and the resonant response is not monotonic. Forward integrations with an explicit evolving mass and size are needed to turn this diagnostic into a constraint.

5.5 What is still not modelled in class 3

Anything before the bar formed (> 8 Gyr ago): GSE’s own infall and disruption (fiducial: $M_\star = 5 \times 10^8$, $M_{\text{DM}} = 2 \times 10^{11} M_\odot$, $c = 4$, from R_{vir} at $z \approx 2$, circularity 0.5, retrograde; [Naidu et al., 2021](#)) and where ω Cen’s host sat in it. Two readings of “deposited by GSE” remain open — ω Cen is GSE’s own nucleus ([Pfeffer et al., 2021](#)), or its host was a smaller dwarf that fell in with GSE (group infall, the oMEGACat X reading) — and both start the disruption on the picked debris

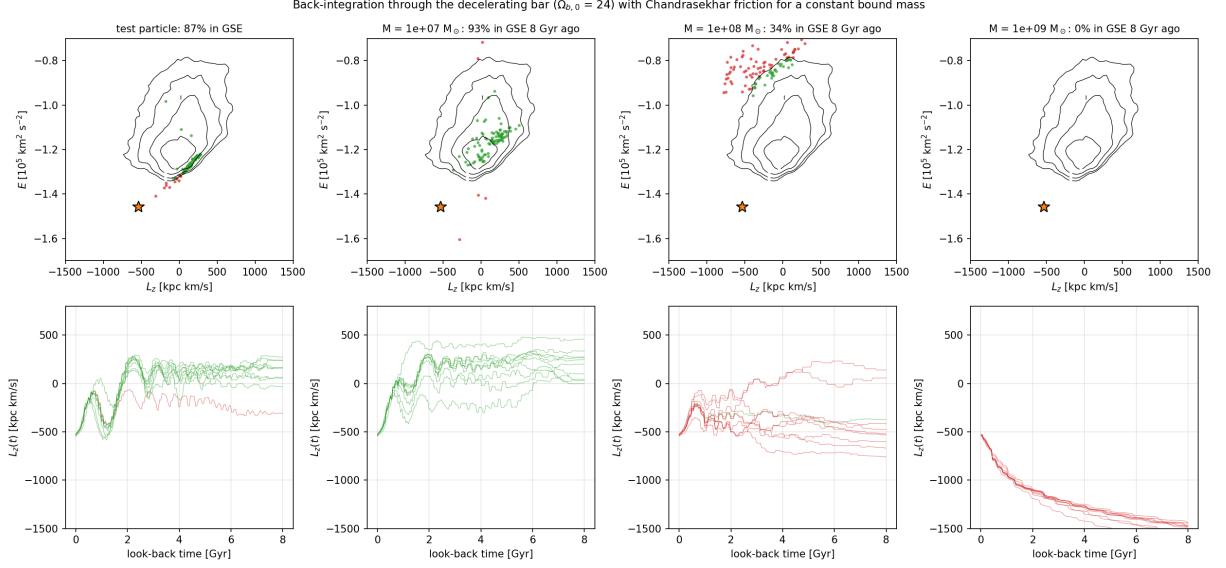


Figure 7: Massive-remnant test at $\Omega_b = 24$. Top: (L_z, E) of the samples 8 Gyr ago for $M = 0, 10^7, 10^8, 10^9 M_\odot$ (green inside, red outside the outer GSE contour). Bottom: $L_z(t)$ of every tenth sample. At $10^7 M_\odot$ the resonance still captures and carries many samples, with shifted endpoints. At $10^8 M_\odot$ the fraction inside the GSE contour is 0.34, compared with 0.93 at $10^7 M_\odot$; at $10^9 M_\odot$ it is zero. Contour membership is an endpoint diagnostic, not by itself proof of resonance capture.

orbit. The bar history is a one-parameter family at fixed η ; the oCen_bar journal notes that the low- Ω_b requirement is tied to this slowdown law, since $\Omega_b = 35$ today implies an implausible $\Omega_1 = 110 \text{ km s}^{-1} \text{ kpc}^{-1}$.

5.6 Which pattern speeds put today’s orbit in resonance?

A different question from “which Ω_b carries ω Cen from GSE to its present (E, L_z) ” is “for which Ω_b is the *present* orbit resonant”. From AGAMA’s action finder in the axisymmetrised Hunter et al. (2024) potential (1000 error samples; `tails/resonances.py`) the present orbit has $\Omega_r = 69.7$, $\Omega_z = 63.1$, $\Omega_\phi = -44.3 \text{ km s}^{-1} \text{ kpc}^{-1}$ (retrograde; McMillan 2017 gives 70.2, 62.8, -44.0), with actions $J_r = 242$, $J_z = 120$, $L_z = -533 \text{ kpc km s}^{-1}$. A resonance with a prograde bar requires

$$l\Omega_r + n\Omega_z + m(\Omega_\phi - \Omega_b) = 0 \quad \implies \quad \Omega_b = \Omega_\phi + \frac{l\Omega_r + n\Omega_z}{m}, \quad (3)$$

and because $\Omega_\phi < 0$ only combinations with enough radial or vertical frequency give a positive Ω_b (Table 7). The sample spread is $\pm 0.1 \text{ km s}^{-1} \text{ kpc}^{-1}$; the potential difference ± 0.5 .

With a bar at the mainstream $33\text{--}41 \text{ km s}^{-1} \text{ kpc}^{-1}$ the present orbit is in no low-order resonance (the nearest is the weak 5:4 at 43). The only strong resonance it can occupy is the retrograde 1:1 at $\Omega_b = 25.4$, which is why the migration scenario needs that pattern speed and, equally, why a fast bar merely perturbs today’s orbit (Sec. 5.7).

5.7 Class 2 in the barred potential

The three class-2 histories were re-integrated in Hunter et al. (2024) — axisymmetrised, and with the growing, decelerating bar ending at the mainstream $37.5 \text{ km s}^{-1} \text{ kpc}^{-1}$ (Hunter’s value) and at 24 — with 20 error samples each (`tails/class2_barred.py`, Fig. 8). The bar is 8 Gyr old; before that the potential is axisymmetric.

With the fast bar nothing changes beyond the sample scatter: the class-2 picks stand. With the slow bar the last ~ 2.5 Gyr become class-3-like (pericentre 1.3–1.7 kpc, apocentre 9–12), because

Table 7: Bar pattern speeds for which ω Cen’s present orbit is resonant.

resonance (l, n, m)	Ω_b [$\text{km s}^{-1} \text{kpc}^{-1}$]	comment
corotation (0, 0, 1)	−44.3	impossible for a retrograde orbit
OLR-type (1, 0, 2)	−9.5	impossible
(3, 0, 4)	8.0	far too slow
vertical (0, 1, 1)	18.7	
(1, 1, 2)	22.1	
retrograde 1:1 (1, 0, 1)	25.4	the Dillamore et al. (2026) resonance
(5, 0, 4)	42.8	high order, weak
(4, 0, 3)	48.6	
(3, 0, 2)	60.3	
(1, 1, 1)	88.5	

Table 8: Class 2 in the barred potential (kpc; medians over samples and over the three histories’ ranges).

	last Gyr peri / apo	5–6 Gyr ago peri / apo	apocentre before infall
axisymmetric	1.6–1.8 / 7.0–7.3	1.8–3.2 / 7–19	64–117
bar, $\Omega_b = 37.5$	1.9–2.3 / 7.1–7.2	1.8–3.4 / 8–19	75–112
bar, $\Omega_b = 24$	1.3–1.7 / 9.1–11.6	1.0–1.9 / 11–28	91–118

the retrograde 1:1 resonance sits at exactly this pattern speed for today’s orbit; the infall apocentre rises by $\sim 20\%$ but stays inside the 50–150 kpc selection band, so the same (M_{inf}, τ) picks remain valid. Under the slow bar the pericentres differ from the axisymmetric case all the way back to infall; that is not the bar acting 9 Gyr ago but a property of back-integration — the whole inferred past is conditioned on today’s state, so a resonant kick in the last 2.5 Gyr re-routes the entire earlier trajectory.

6 Comparison between the classes

Figure 10 and Table 9 summarise what a disruption simulation would see.

Table 9: Per-Gyr elements at three epochs (kpc unless stated). Tidal strength $T \equiv GM(< r_{\text{peri}})/r_{\text{peri}}^3$ in $10^3 (\text{km s}^{-1}/\text{kpc})^2$; r_J is the Jacobi radius of a $10^9 M_\odot$ host at pericentre in the last Gyr.

class	orbit	peri (0–1 / 5–6 / 8–9 Gyr)	apo (0–1 / 5–6 / 8–9)	T (0–1)	T (8–9)	$r_J^{10^9}$
1	McMillan 2017	1.59 / 1.58 / 1.58	7.0 / 7.0 / 7.0	14.9	14.9	0.46
1	MWPotential2014	1.98 / 1.98 / 1.97	6.8 / 6.8 / 6.8	10.3	10.3	0.52
1	Irrgang 2013 I	1.25 / 1.26 / 1.25	7.2 / 7.2 / 7.2	21.7	21.7	0.40
2	Sequoia-like	1.60 / 2.95 / 5.18	7.4 / 18.1 / 48.2	14.7	1.8	0.46
2	intermediate	1.59 / 2.24 / 5.36	7.1 / 11.7 / 43.1	14.8	1.7	0.46
2	GSE-like	1.58 / 1.69 / 4.54	7.0 / 7.4 / 39.5	14.9	2.3	0.46
3	sample 986	0.84 / 0.59 / 0.40	9.1 / 11.0 / 10.9	39.4	105.2	0.33
3	sample 133	0.80 / 0.71 / 0.42	9.3 / 11.7 / 11.6	42.0	98.6	0.32
3	sample 240	0.84 / 0.80 / 0.45	9.1 / 11.1 / 12.2	39.4	90.0	0.33

Late history (last 3 Gyr). Classes 1 and 2 are indistinguishable: pericentre 1.6 kpc, apocentre 7 kpc, $T \approx 1.4 \times 10^4 (\text{km s}^{-1}/\text{kpc})^2$. Class 3 is different even today because the slow bar is part

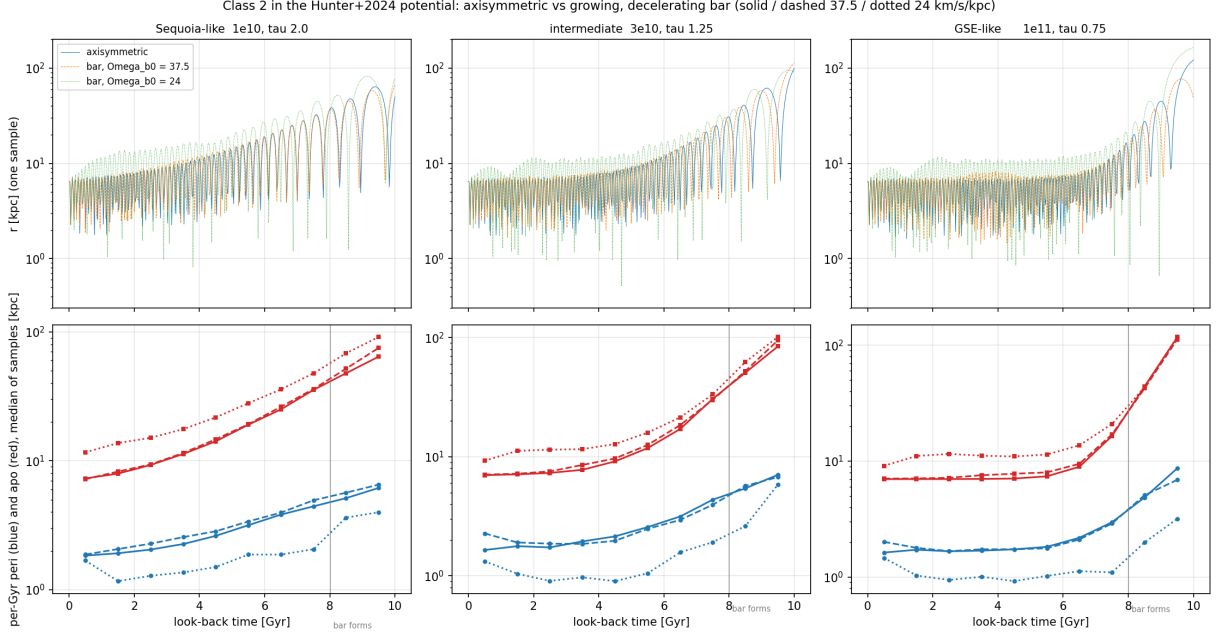


Figure 8: Class 2 in the Hunter et al. (2024) potential: axisymmetric (solid), bar ending at 37.5 (dashed) and at $24 \text{ km s}^{-1} \text{ kpc}^{-1}$ (dotted). Top: one sample’s radius. Bottom: per-Gyr pericentre (blue) and apocentre (red), median over 20 samples; the bar forms 8 Gyr ago.

of its potential: pericentre 0.8 kpc and three times stronger pericentric tides. Whether that is a property of class 3 or of the Galaxy depends only on whether the bar is slow; if it is, classes 1 and 2 should be re-evaluated in the barred potential as well.

Early history (5–10 Gyr ago). This is where the classes separate, by more than an order of magnitude in tidal strength. Class 1 keeps today’s tides throughout. Class 2 has a gentle early history: pericentres of $2\text{--}6 \text{ kpc}$ and T ten times weaker than today, so the host is stripped slowly at first and hardest at the end, when it is already light. Class 3 is the opposite: pericentres of 0.4 kpc through the bulge with T seven times *stronger* than today and a Jacobi radius of only 0.3 kpc for a $10^9 M_\odot$ host — the host is destroyed early and violently, and what survives to be migrated by the bar is the nucleus alone. For the dark-matter question this is the sharpest distinction: a class-3 history leaves the nucleus the least time to retain a dark envelope, a class-2 history the most.

Angular momentum. The early orbit is more retrograde than today in class 2 ($L_z \rightarrow -2000$ to $-4000 \text{ kpc km s}^{-1}$ as friction is undone) and *less* retrograde in class 3 ($L_z \approx 0$ to $+230$; Fig. 11). Retrograde class-2 infall is consistent with the retrograde Sequoia/Thamnos association; a class-3 host arrives on the near-radial GSE orbit.

Robustness. Class 1 depends only on the potential (factor 1.6 in pericentre). Class 2 depends on $M(t)$ far more than on any single mass, and the selection rule of Sec. 4.2 is what makes the set finite; the friction formula itself is a 20–30% approximation. Class 3 depends on the bar’s slowdown law and on a present-day pattern speed at the low end of the literature, and its success is a jagged function of Ω_b .

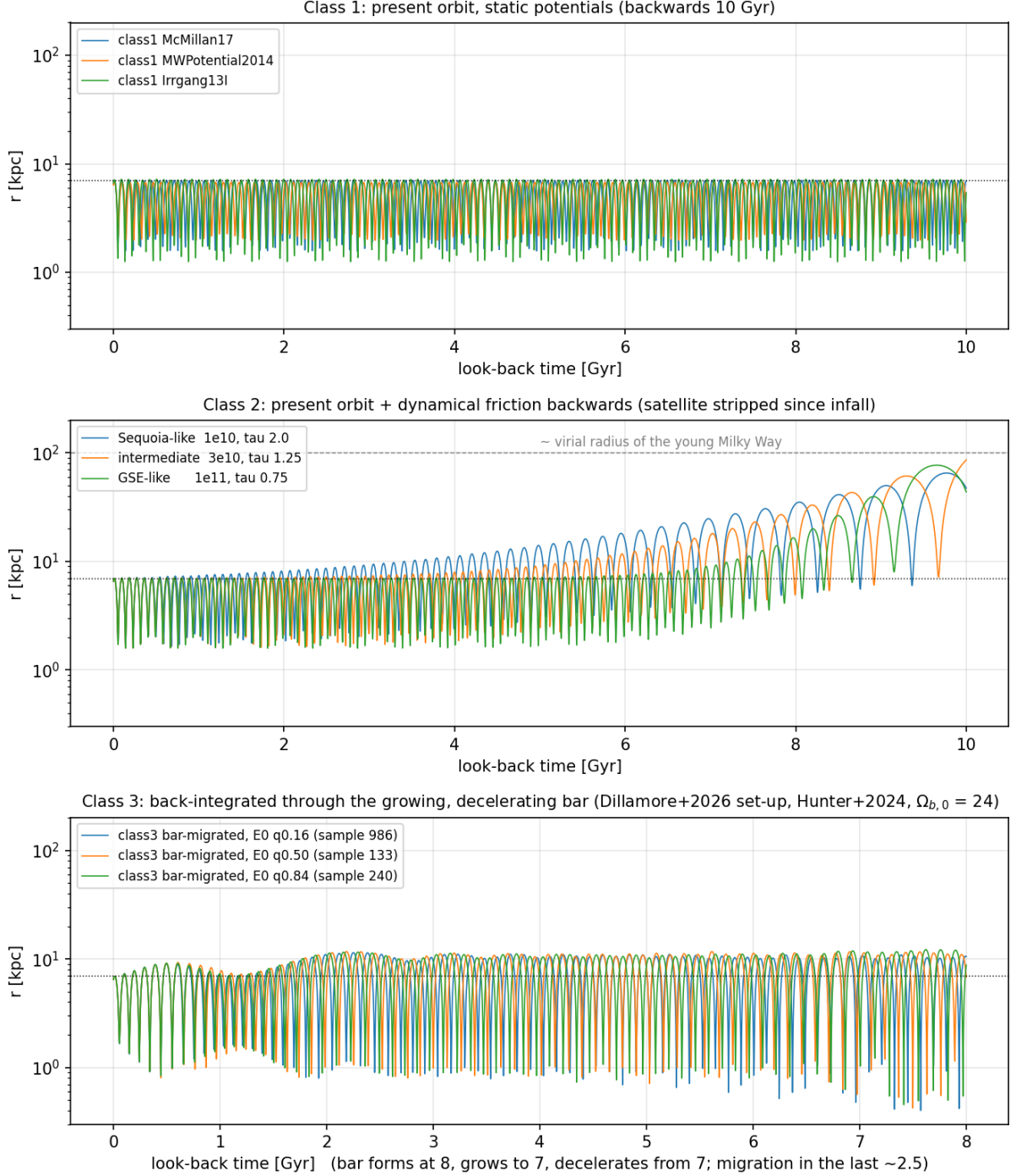


Figure 9: Overview of the nine orbits: galactocentric radius against look-back time for classes 1 (top), 2 (middle; dashed line 100 kpc) and 3 (bottom). Dotted line: today's apocentre, 7 kpc.

7 Tidal tracks: what the orbits imply before any N -body run

7.1 The method and why it separates structure from orbit

Peñarrubia et al. (2010) and Errani & Navarro (2021, EN21) showed that the structure of a tidally stripped NFW subhalo depends only on how much mass it has lost, not on how: the characteristic velocity and radius follow one track,

$$\frac{V_{\text{mx}}}{V_{\text{mx},0}} = 2^\alpha \left(\frac{r_{\text{mx}}}{r_{\text{mx},0}} \right)^\beta \left[1 + \left(\frac{r_{\text{mx}}}{r_{\text{mx},0}} \right)^2 \right]^{-\alpha}, \quad \alpha = 0.4, \beta = 0.65, \quad (4)$$

The associated remnant density is an exponentially truncated cusp, $\rho = \rho_{\text{NFW}} e^{-r/r_{\text{cut}}}/(1 + r_s/r_{\text{cut}})^{0.3}$ with $r_{\text{cut}}/r_{\text{mx},0} = 0.44 (M_{\text{mx}}/M_{\text{mx},0})^{0.44} [1 - (M_{\text{mx}}/M_{\text{mx},0})^{0.3}]^{-1.1}$. The orbit enters only through the speed along the track: with the crossing time $T_{\text{mx}} = 2\pi r_{\text{mx}}/V_{\text{mx}}$ and the circular time at pericentre $T_{\text{peri}} = 2\pi r_{\text{peri}}/V_c$, EN21 find that the remnant converges to $T_{\text{asy}} \simeq 0.22 T_{\text{peri}}$, i.e. to a mean density $\simeq 20$ times the host’s mean density inside the pericentre, along $Y(t) = (T_{\text{mx}} - T_{\text{asy}})/T_{\text{peri}} = Y_0[1 + (t/\tau)^\eta]^{-1/\eta}$ with $\tau = 0.65 T_{\text{orb}}/Y_0$, $\eta = 1 - e^{-2.5Y_0}$, and eccentric orbits delayed by $f_{\text{ecc}} = [2x/(x+1)]^{3.2}$, $x = r_{\text{apo}}/r_{\text{peri}}$. Because the state is a single number on the track, a time-varying orbit is handled passage by passage with the local $(T_{\text{peri}}, T_{\text{orb}}, f_{\text{ecc}})$ (`tails/tidal_tracks.py`, `walk`); the host mean density inside each pericentre replaces the isothermal T_{peri} (EN21’s own suggested generalisation, with their caveat).

7.2 Our orbits are outside the calibrated regime

EN21 simulated $0.2 < T_{\text{mx},0}/T_{\text{peri}} < 2$. For NFW haloes of 10^9 – $10^{11} M_\odot$ at $z = 2$ ($c = 4.5$ – 7 ; $r_{\text{mx},0} = 3$ – 22 kpc, $T_{\text{mx},0} = 0.8$ – 1.4 Gyr) on our orbits the ratio is $T_{\text{mx},0}/T_{\text{peri}} = 8$ – 65 : the pericentric tides are so much stronger than anything in the calibration that the walk extrapolates to remnants of $M_{\text{mx}}/M_{\text{mx},0} \sim 10^{-12}$ within a few passages — not calibrated predictions. The asymptotic density criterion provides a diagnostic scale rather than a rigorous survival threshold in this regime: it gives $\simeq 20 \bar{\rho}_{\text{host}}(< r_{\text{peri}})$, which is 1.7×10^{10} (classes 1–2, $r_{\text{peri}} = 1.6$ kpc) to $1.5 \times 10^{11} M_\odot \text{ kpc}^{-3}$ (class 3, 0.4 kpc). An NFW cusp of $10^{10} M_\odot$, $c = 5.5$ has that mean density only inside 4.6 pc ($6.7 \times 10^3 M_\odot$) for classes 1–2 and inside 0.5 pc ($\sim 10^2 M_\odot$) for class 3; a concentrated $c = 12$ halo inside 14 pc (1.7×10^5) and 1.6 pc (2.4×10^3). These small diagnostic scales motivate including the nucleus’s gravity explicitly. They do not establish the mass left by a full disruption simulation. Cored haloes are more vulnerable in the calibrated models (Errani & Peñarrubia, 2020); extrapolation to these histories still needs validation.

7.3 The dark-matter budget the nucleus can hold

That reframes the question as a budget: the dark matter initially inside the nucleus’s Jacobi radius at the smallest pericentre, $r_J = r_{\text{peri}}[M_{\text{nuc}}/3M_{\text{host}}(< r_{\text{peri}})]^{1/3}$, is an initial spatial budget, not a computed retained mass (Table 10, Fig. 12).

Table 10: Initial NFW dark matter inside the nucleus’s Jacobi radius at the minimum pericentre ($M_\odot$); range over the orbits of each class. Nucleus mass 3.55×10^6 .

	r_J [pc]	$10^9, c = 7$	$10^{10}, c = 5.5$	$10^{11}, c = 4.5$	/	$10^{10}, c = 12$
class 1	40–70	$3\text{--}9 \times 10^5$	$0.5\text{--}1.5 \times 10^6$	$0.8\text{--}2.6 \times 10^6$	/	$1.5\text{--}4.4 \times 10^6$
class 2	70	9×10^5	1.5×10^6	2.6×10^6	/	4.4×10^6
class 3	33–36	$2.2\text{--}2.5 \times 10^5$	$3.5\text{--}4.0 \times 10^5$	$5.9\text{--}6.7 \times 10^5$	/	$1.0\text{--}1.2 \times 10^6$

The initial aperture budgets differ between classes and can be comparable to the nuclear mass. They omit energy and angular-momentum selection, self-consistent re-adjustment, contraction,

shocks and stellar heating. They are not rigorous upper limits for arbitrary formation histories, and DM self-gravity is not negligible for every grid point.

A smaller experiment can focus on the inner ~ 100 pc using test particles or a live DM component around an extended nucleus. This is a proposed reduction, not a consequence already validated by the tidal tracks. Particles starting farther out can pass through the centre, and an imposed loss of the outer potential needs explicit justification. The current validation sequence is in `docs/SECTION11_TESTS.md`; the companion WD note distinguishes an illustrative spatial taper from a calculated truncated DF.

8 Caveats and reproducibility

- Corrections after the Codex review of 2026-09-20: the leapfrog unit bug above; the class-3 tidal-radius proxy at the 0.4 kpc pericentre is about 33 pc. A historical alternative estimate of 59 pc has not been reproduced by a consistent tidal-tensor calculation and is not adopted. For circular spherical orbits the denominator coefficient is $3 - d \ln M / d \ln R$; eccentric barred orbits need an explicitly stated approximation; two of the class-2 stripping histories end today at 6.7×10^7 and $1.0 \times 10^7 M_\odot$ rather than at the nucleus mass (a constraint worth imposing in a revision).
- Chandrasekhar friction with $\sigma = v_c / \sqrt{2}$, a fixed r_h and a smooth exponential stripping law; no orbit-phase-dependent mass loss.
- Only McMillan (2017) for class 2; only Hunter et al. (2024) for class 3 (the only potential with a fitted bar); class 1 carries the potential spread.
- Tidal proxies in Fig. 10 use McMillan (2017) enclosed masses for all classes.
- The `galpy` checkout used (2024-03) needs two import shims (`astroquery metadata`, `scipy vectorize1`); both live in `progenitor_orbits._import_galpy_safely`.
- Products: `python-mocen_dm.tails.products`, `python-mocen_dm.tails.bar_migration`, `python-mocen_dm.tails.writeup_figures`. The class-3 run of 14×10^3 orbits takes about a minute.

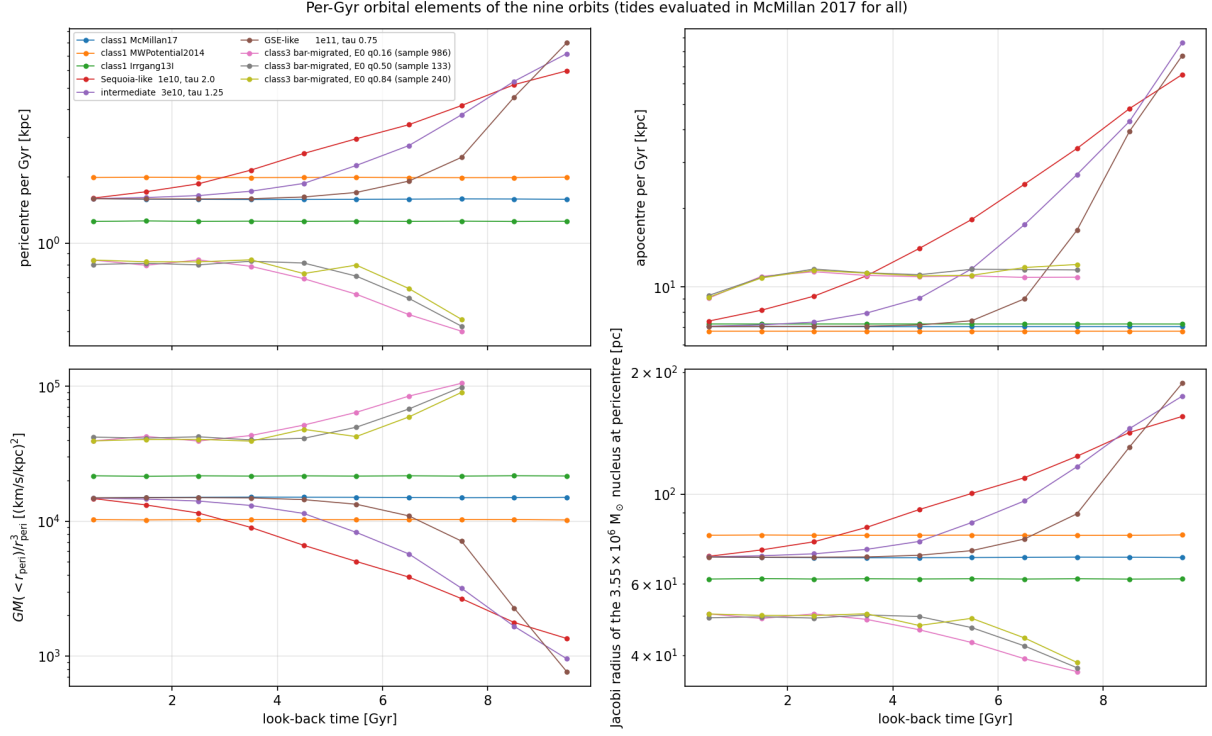


Figure 10: Per-Gyr orbital elements of the nine orbits: pericentre, apocentre, tidal-field strength $GM(< r_{\text{peri}})/r_{\text{peri}}^3$ at pericentre, and the Jacobi radius a $3.55 \times 10^6 M_{\odot}$ nucleus would have there. Enclosed masses from McMillan (2017) for all orbits, including class 3 (integrated in Hunter et al. 2024), for a uniform proxy.

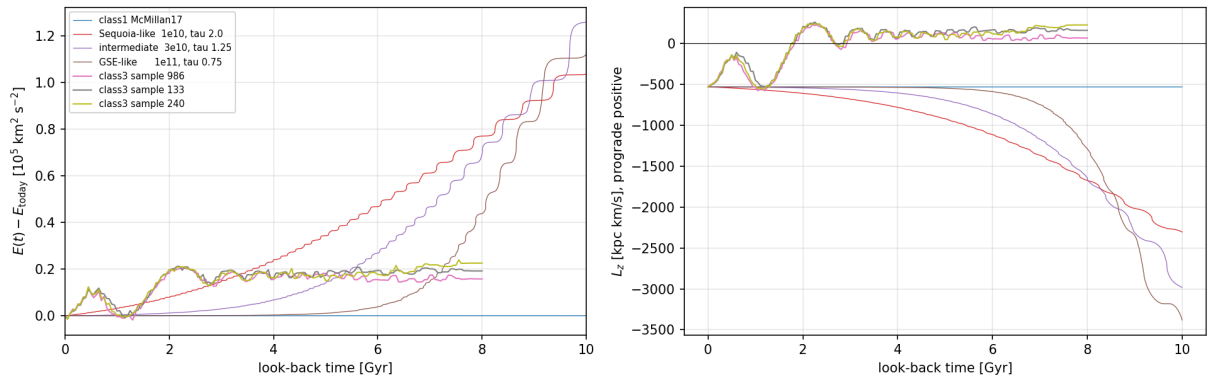


Figure 11: Energy relative to today (left; each orbit in its own host potential) and L_z (right, prograde-positive) against look-back time. Class 1 (McMillan 2017 only) is flat by construction; class 2 climbs steadily as friction is undone; class 3 is flat at GSE values until the resonance moves it in the last 2.5 Gyr. Note the opposite directions of the L_z change: friction makes the early orbit *more* retrograde, the bar makes it *less*.

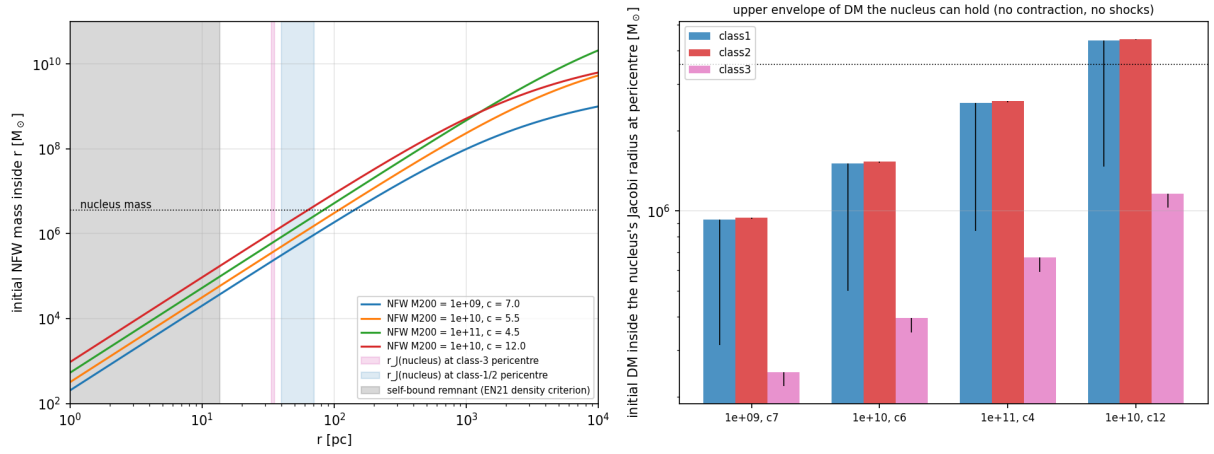


Figure 12: Left: initial NFW mass profiles at $z = 2$ with the self-bound-remnant scale from the EN21 density criterion (grey) and the nucleus's Jacobi radius at the class-3 (pink) and class-1/2 (blue) pericentres. Right: the dark matter initially inside that Jacobi radius, by class and halo. These are initial aperture masses under the stated tidal-radius proxy.

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